

PROJECT

5.1

**BUILD A VIRTUAL
PRIVATE CLOUD
NETWORKING SERIES**

Prepared by:
Amrize JK
Certified Solutions Architect Associate

Objective

Amazon VPC helps you build a private and secure network in the cloud. It creates a safe space where your applications and resources stay isolated from the public internet.

With VPC, you have full control over how your network works. You can manage IP addresses, subnets, routing rules, and gateways to decide what connects to the internet and what stays private.

Services

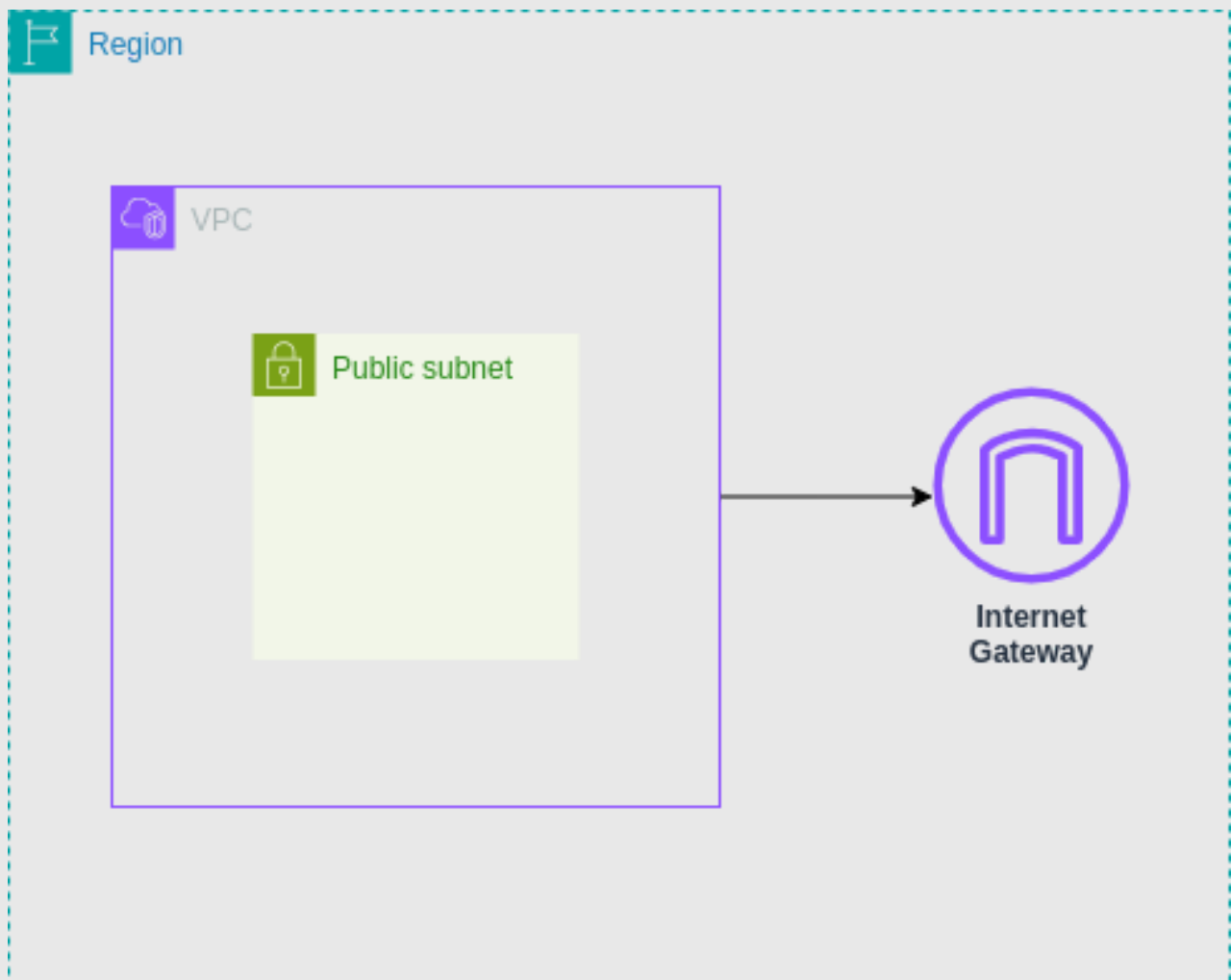
Amazon VPC - Creates an isolated, configurable network environment in the AWS Cloud.

Subnets (Public/Private) - Divide your VPC to control access between public and internal resources.

Internet Gateway - Provides internet connectivity for resources in public subnets.

AWS CloudShell - A browser-based command line tool to manage AWS services without local setup.

Architecture Diagram



In this setup, a Region hosts your cloud network. Inside it, a VPC provides an isolated environment. A Public Subnet within the VPC holds resources. An Internet Gateway connects the subnet to the internet, enabling secure and controlled access.

Steps to Implement

1. Create a VPC:

Start by creating your own Virtual Private Cloud (VPC), which is like setting up your own private network inside AWS. You can define the IP address range using CIDR blocks (like 10.0.0.0/16). This step gives you full control over your cloud networking environment like a secure container for all your AWS resources.

2. Create Subnets:

Once your VPC is ready, create subnets inside it. Subnets are smaller divisions within the VPC to organize and separate resources. For public access, you create a Public Subnet that can connect to the internet. You can also create private subnets for internal only resources if needed.

3. Create an Internet Gateway:

To allow communication between your VPC and the internet, you need to create an Internet Gateway (IGW). This acts like a door between your VPC and the outside world. After creating it, attach the IGW to your VPC so that any public subnet in the VPC can access the internet.

4. Launch a VPC in Seconds with AWS CloudShell:

With all components ready, you can quickly deploy your setup using AWS CloudShell, a browser-based terminal pre configured with AWS CLI. You can run simple commands to launch EC2 instances inside your public subnet, test internet connectivity, and verify your architecture setup within seconds.

Story Time

VPC (Your City Walls): This is like building a strong wall around your city. You decide how big your city should be (by choosing the area with IP addresses). No strangers can come in without your permission!

Subnets (Your Neighborhoods): Inside your city, you build different neighborhoods (subnets). One can be for visitors (public subnet), and another one can be just for your home or secret base (private subnet).

Internet Gateway (Your City Gate): You want to let friendly visitors in or send letters outside, right? The Internet Gateway is like the big main gate of your city. Only neighborhoods connected to this gate can talk to the outside world (internet).

CloudShell (Your Magic Wand): Now, to make all this happen quickly, you use AWS CloudShell, your magic wand! Just type a few commands, and “boom” your city is live and working!

BY
AMRIZE J.K

THANK YOU!

